

Computer Science & IT

Database Management System



Transactions & concurrency control

Lecture No. 07



By- Vishal Sir

Recap of Previous Lecture



CSS



Topic

Precedence graph ✓

Topic

Topological order ✓

Topic

Testing condition for Conflict serializability

Topic

Practice question

Topics to be Covered



✓
Topic

[View Serializable Schedule](#)





View Serializable Schedule



Topic : View serializable schedule

If given schedule is view equivalent to at least one of the serial schedule, then it is called a **view serializable schedule**.

Converse of the statement is also true.

i.e. if schedule is view serializable schedule then it will be view equivalent to at least one of the serial schedule



Topic : View equivalent condition

- Consider two schedules S_1 and S_2
- Schedules S_1 and S_2 are called view equivalent if the following three conditions hold true for them –
- ❖ ☐ **Condition-01** : For each data item X , if transaction T_i reads X from the initial database in schedule S_1 , then in schedule S_2 also, T_i must perform the XXXX read of X from the initial database.

Thumb Rule

- ☐ “Initial readers must be same for all the data items”.



Topic : View equivalent condition

Condition-02 :

If transaction T_i reads the value of data item X updated by the transaction T_j in schedule S_1 , then in schedule S_2 also, transaction T_i must read the value of data item X updated by the transaction T_j .

Thumb Rule

- ❑ "Write-read sequence must be same."



Topic : View equivalent condition

Condition-03 :

For each data item X , if X is finally updated by transaction T_i in schedule S_1 , then in schedule S_2 also, X must be finally updated by transaction T_i .

Thumb Rule

- ❑ “Final writers must be same for all the data items”.

→ For two schedules to be view equivalent all three conditions must hold true

① Q:- Check whether the following schedule is a view serializable schedule or not, if view serializable schedule then identify all "view equivalent serial schedules"

(S₁)

T ₁	T ₂	T ₃
W ₁ (A)	R ₂ (B) R ₂ (A) W ₂ (B)	
W ₁ (B)		W ₃ (B)

Q:- Check whether the following schedule is a view serializable schedule or not, if view serializable schedule then identify all "view equivalent serial schedules"

(S₁)

① Identify the constraints w.r.t. Read from initial DB

T ₁	T ₂	T ₃	Data item	Transaction that reads the dataitem from initial DB	Other transactions that writes that dataitem	Constraints w.r.t. view Equivalent serial schedule
W ₁ (A)	R ₂ (B) R ₂ (A) W ₂ (B)		A	No transaction reads dataitem A from initial DB	T ₁	No Constraint w.r.t read from initial DB
W ₁ (B)		W ₃ (B)	B	T ₂	(T ₁ , T ₃)	T₂ → (T₁, T₃) i.e., T₂ → T₁ T₂ → T₃ { i.e. T₂ must execute before T₁ & T₃ in Equivalent serial schedule }

Q1:- Check whether the following schedule is a view serializable schedule or not, if view serializable schedule then identify all "view equivalent serial schedules"

(S₁)

T ₁	T ₂	T ₃
W ₁ (A)	R ₂ (B) R ₂ (A) W ₂ (B)	
W ₁ (B)		W ₃ (B)

Q2 Identify the constraints w.r.t. Write-read sequence

Concept: Let transaction T_i updates the dataitem X in the given schedule and transaction T_j reads the value of X updated by transaction T_i , also assume that T_k is some other transaction in the schedule which also update the dataitem ' X ' then in any equivalent serial schedule

- ① T_i must execute before T_j
- and ② transaction T_k (if any such transaction exists) must not be allowed to execute in between T_i & T_j

ie, $T_i \rightarrow T_j$

~~T_k~~ not allowed

Q1: - Check whether the following schedule is a view serializable schedule or not, if view serializable schedule then identify all "view equivalent serial schedules"

(S₁)

② Identify the constraints w.r.t. Write-read sequence
Constrain w.r.t Write-read sequence

T ₁	T ₂	T ₃
W ₁ (A)	R ₂ (B) R ₂ (A) W ₂ (B)	
W ₁ (B)		W ₃ (B)

Data item

A :

T₂ reads the value of dataitem 'A' updated by T₁,
{ No other writer of dataitem A in the schedule
ie. T_k = ∅
∴ Constraint will be T₁ → T₂

No other writer of A
∴ No Constraints w.r.t. T_k

B

No Write-Read sequence
w.r.t. dataitem 'B' ∴ No Constraints w.r.t. dataitem 'B'

① Q:- Check whether the following schedule is a view serializable schedule or not, if view serializable schedule then identify all "view equivalent serial schedules"

(S₁)

③ Identify the constraints w.r.t. Final Write

T ₁	T ₂	T ₃
W ₁ (A)	R ₂ (B) R ₂ (A) W ₂ (B)	
W ₁ (B)		W ₃ (B)

Data item	Transaction with the final update of data item	Other transactions that update the data item	Constraints w.r.t. Equivalent serial schedule
A:	T ₁	No other transaction updates A.	No constraints w.r.t. data item A
B	T ₃	(T ₁ , T ₂)	(T ₁ , T ₂) → T ₃ { i.e. T ₁ & T ₂ must execute before T ₃ i.e. T₁ → T₃ T₂ → T₃

Note :- Once we have identified the Constraints w.r.t. all three conditions construct the dependency graph w.r.t. overall Constraints

- If dependency graph is cyclic then schedule is not a view serializable schedule, and hence the schedule is a non-serializable schedule
- If dependency graph is acyclic then schedule is a view serializable schedule, and hence the schedule is a view serializable schedule

① Q:- Check whether the following schedule is a view serializable schedule or not, if view serializable schedule then identify all "view equivalent serial schedules"

(S₁)

T ₁	T ₂	T ₃
W ₁ (A)	R ₂ (B) R ₂ (A) W ₂ (B)	
W ₁ (B)		W ₃ (B)

→ Set of overall constraints identified

T₂ → T₁
T₂ → T₃ } Wrt read from initial DB

T₁ → T₂ } Wrt. write-read sequence

T₁ → T₃
T₂ → T₃ } Wrt. Final write

In the overall constraints if we find two constraints s.t. T_i → T_j and T_j → T_i, then dependency graph is guaranteed to be cyclic, and we can say that schedule is not a view serializable schedule, and hence not a serializable schedule

Q1:- Check whether the following schedule is a view serializable schedule or not, if view serializable schedule then identify all "view equivalent serial schedules"

S₁

T ₁	T ₂	T ₃
W ₁ (A)	R ₂ (B) R ₂ (A) W ₂ (B)	
W ₁ (B)		W ₃ (B)

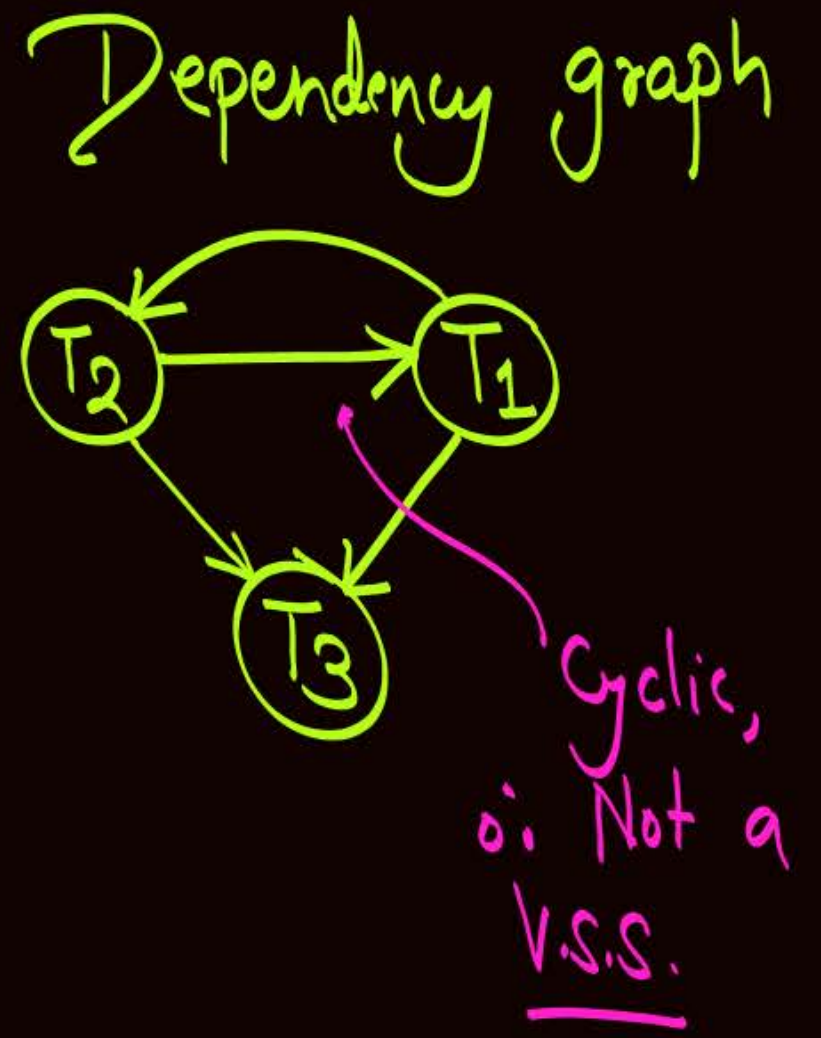
→ Set of overall constraints identified

$T_2 \rightarrow T_1$
 $T_2 \rightarrow T_3$

$T_1 \rightarrow T_2$

$T_1 \rightarrow T_3$
 $T_2 \rightarrow T_3$

} Wrt read from initial DB
 } Write-read sequence
 } Wrt. Final write



② Q:- Check whether the following schedule is a view serializable schedule or not, if view serializable schedule then identify all "view equivalent serial schedules"

(S₂)

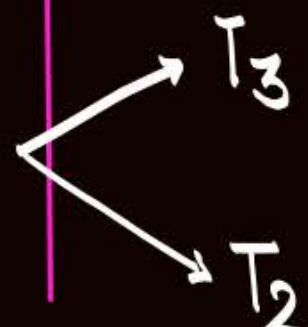
T ₁	T ₂	T ₃
		R ₃ (B)
W ₁ (A)	R ₂ (B)	
	R ₂ (A)	
	W ₂ (B)	
W ₁ (B)		
		W ₃ (B)

② Q:- Check whether the following schedule is a view serializable schedule or not, if view serializable schedule then identify all "view equivalent serial schedules"

(S2)

① Identify the constraints w.r.t. Read from initial DB

T ₁	T ₂	T ₃
W ₁ (A)	R ₂ (B) R ₂ (A) W ₂ (B)	R ₃ (B)
W ₁ (B)		W ₃ (B)

Data Item	Transaction that reads the data item from initial DB	Other transactions that writes the data item	Constraints w.r.t. View Equivalent Serial Schedule
A	—	T ₁	No Constraint
B		(T ₁ , T ₂) (T ₁ , T ₃)	$T_3 \rightarrow (T_1, T_2) \equiv \begin{matrix} T_3 \rightarrow T_1 \\ T_3 \rightarrow T_2 \end{matrix}$ $T_2 \rightarrow (T_1, T_3) \equiv \begin{matrix} T_2 \rightarrow T_1 \\ T_2 \rightarrow T_3 \end{matrix}$

$T_i \rightarrow T_j$
 $T_j \rightarrow T_i$
 $\therefore$ Cycle.
 Hence Not a V.S.S.

② Identify the constraints w.r.t. Write-Read Seq
 {Exactly same as previous question}

③ Identify the constraints w.r.t. Final-Write
 {Exactly same as previous question}

③ Q:- Check whether the following schedule is a view serializable schedule or not, if view serializable schedule then identify all "view equivalent serial schedules"

S ₃	
T ₁	T ₂
W ₁ (A)	
	R ₂ (A)
	W ₂ (B)
R ₁ (B)	

① Constraints w.r.t. read from initial DB: -

Dataitem $\begin{cases} A : \text{No Constraints} \\ B : \text{No Constraints} \end{cases}$

② Constraints w.r.t. write read seq

Dataitem $\begin{cases} A : T_1 \rightarrow T_2 \\ B : T_2 \rightarrow T_1 \end{cases}$

$T_k = \emptyset : \therefore \text{No Constraint w.r.t. } T_k$

$T_k = \emptyset : \therefore \text{No Constraint w.r.t. } T_k$

$T_i \rightarrow T_j \neq$
 $T_j \rightarrow T_i$

$\therefore \text{Not a V.S.S.}$

③ Constraints w.r.t. final write

Dataitem $\begin{cases} A : \text{No Constraints} \\ B : \text{No Constraints} \end{cases}$

④ Q:- Check whether the following schedule is a view serializable schedule or not, if view serializable schedule then identify all "view equivalent serial schedules"

(S₄)

T ₁	T ₂	T ₃
	R(B)	
	R(A)	
W(A)		
	W(A)	
		W(A)

① Identify the Constraints w.r.t. Read from initial DB

A: $T_2 \rightarrow (T_1, T_3) \equiv \begin{matrix} T_2 \rightarrow T_1 \\ T_2 \rightarrow T_3 \end{matrix}$

B: No Constraints

② Identify the Constraints w.r.t. Write-Read Seq

A: No W-R Seq; $\therefore$ No Constraints

B: No WR Seq; $\therefore$ No Constraints

③ Identify the Constraints w.r.t. Final-Write

A: $(T_1, T_2) \rightarrow T_3 \equiv \begin{matrix} T_1 \rightarrow T_3 \\ T_2 \rightarrow T_3 \end{matrix}$

B: No Constraints

→ Overall Constraints

$$\begin{matrix} T_2 \rightarrow T_1 \\ T_2 \rightarrow T_3 \\ T_1 \rightarrow T_3 \end{matrix}$$

→ Dependency graph



Acyclic dependency graph,
 $\therefore$ Schedule S₄ is a view serializable schedule
 And View Equivalent serial schedule is $T_2 \rightarrow T_1 \rightarrow T_3$

#Q. Check whether the schedule is view serializable schedule or not?

If view serializable schedule then identify all view equivalent serial schedules.

T1	T2	T3	T4
R(A)			
	R(A)		
		R(A)	
			R(A)
W(B)			
	W(B)		
		W(B)	
			W(B)

① Constraints Writ. read from initial DB :

Dataitem $\begin{cases} A: \text{No Constraints} \\ B: \text{No Constraints} \end{cases}$

② Constraints Writ. W-R seq

Dataitem $\begin{cases} A: \text{No Constraints} \\ B: \text{No Constraints} \end{cases}$

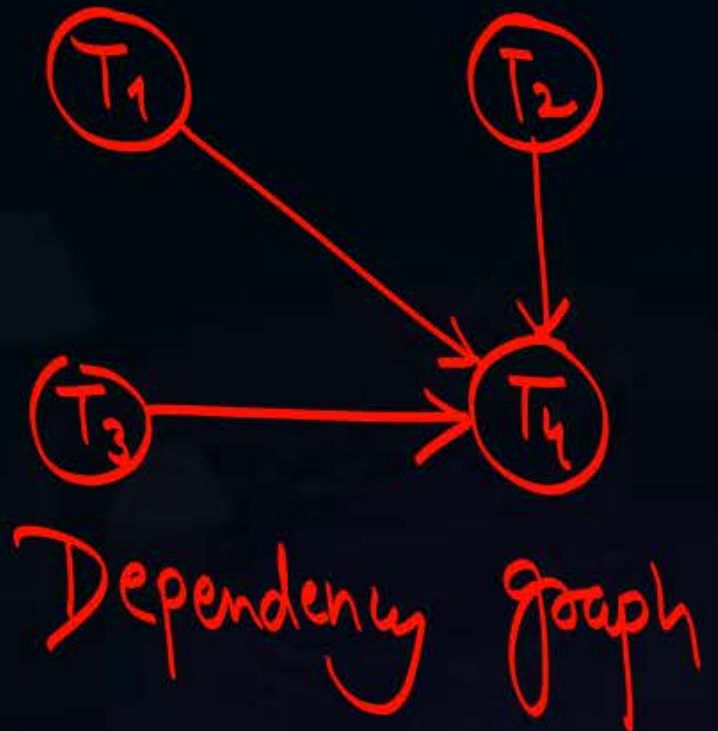
③ Constraints Writ. final write

A: No Constraints

B: $(T_1, T_2, T_3) \rightarrow T_4$

it is the only constraint

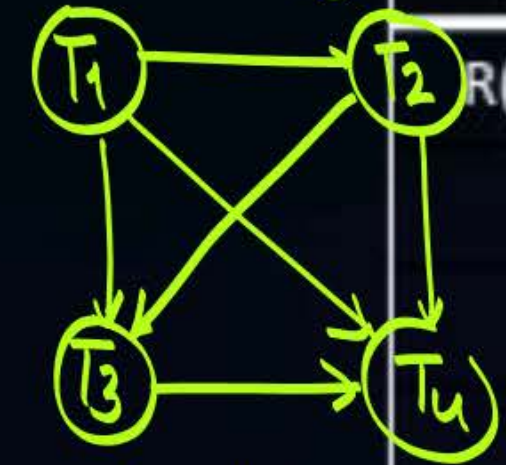
$\therefore 3! \text{ ways to execute } (T_1, T_2, T_3) \text{ before } T_4, \therefore 6 \text{ View equivalent serial schedules}$



#Q. Check whether the schedule is view serializable schedule or not?

If view serializable schedule then identify all view equivalent serial schedules.

Precedence graph of S



Acyclic
Topological order

$T_1 \rightarrow T_2 \rightarrow T_3 \rightarrow T_4$

only one topological order

∴ Only one Conflict equivalent serial schedule.

⑤

T1	T2	T3	T4
R(A)			
	R(A)		
		R(A)	
			R(A)
W(B)			
	W(B)		
		W(B)	
			W(B)

How many serial schedules are there that are view equivalent to schedule 'S' but not Conflict equivalent to schedule S.

Ans = No. of serial schedules that are view equivalent to schedule S

No. of serial schedules that Conflict equivalent to schedule S

$$= 6 - 1 = 5 \text{ Ans}$$

#Q. Check whether the schedule is view serializable schedule or not?

If view serializable schedule then identify all view equivalent serial schedules.

T1	T2	T3
R(A)		
	R(B)	
	W(B)	
		R(A)
		W(A)
R(B)		
		R(B)
	R(A)	
	W(A)	

① initial Read
 A: $T_1 \rightarrow (T_2, T_3)$
 $T_3 \rightarrow T_2$

B: No Constraints

② W-R seq

A: $T_3 \rightarrow T_2$

$T_k = \emptyset \therefore$ No other Constraints

B: $T_2 \rightarrow T_1$

$T_2 \rightarrow T_3$

③ Final write

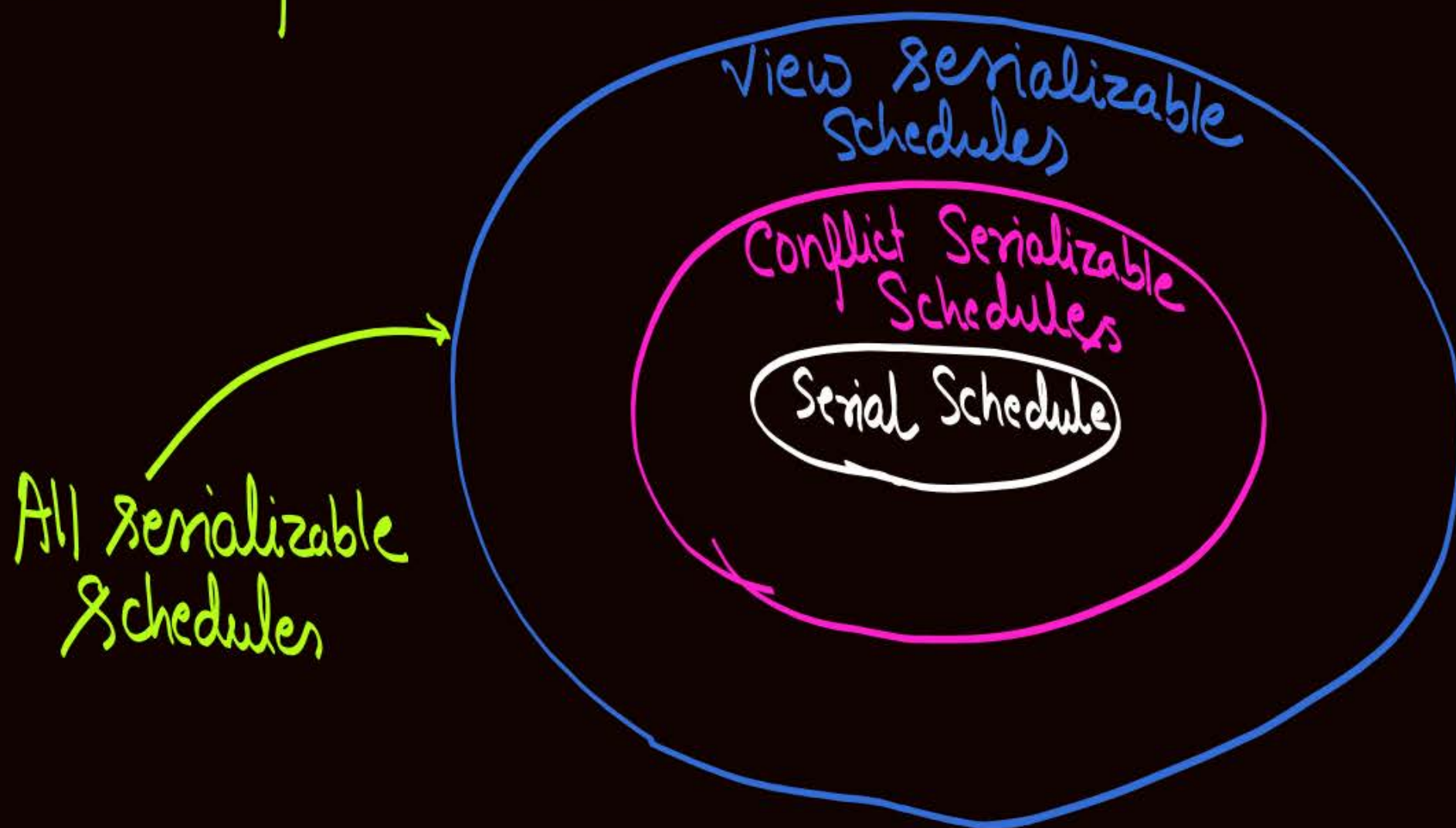
A: $T_3 \rightarrow T_2$

B: No Constraints

Cycle is Not a V.S.S.

Note:

Every Conflict Serializable Schedule is a View Serializable Schedule, but Every View Serializable Schedule need not be Conflict Serializable Schedule





2 mins Summary



Topic

View Serializable Schedule

Slide



THANK - YOU